

SURV 621/SURVMETH 621 Fundamentals of Data Collection

In-class exercise: Calculating survey response rates

Monday, September 9, 2024

You have been given a file containing final disposition codes for a household survey sample. According to the information you have, the number of cases in each of the categories you have been provided is as follows:

500	Completed interviews
100	Partial completes
900	Eligible cases without a completed interview
1,000	Cases of unknown eligibility
500	Not eligible

Answer the following questions related to the calculation of AAPOR response rates for the survey.

- 1) What is RR1 in this sample?
- 2) What is RR2 in this sample?
- 3) If you assume that the share of cases of unknown eligibility who would have been eligible is the same as the share among the cases whose eligibility was determined, what is your estimate of e ?
- 4) Given the estimate of e from 3), what is RR3 in this sample?
- 5) What is the estimate of RR5 in this sample?

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You are conducting a survey to learn about the earnings of IT professionals working in a large metropolitan area. The target population for the survey consists of all employers in the area and is well-captured by the list of businesses you are using as your sampling frame. You have stratified your sample into four size classes based on the total number of employees at each establishment. This same information allows you to calculate average employment among the establishments in each size class. The number of establishments by size class on the frame, in the sample, and among those who respond to your survey is shown in the table below:

Range in number of employees	Average number of employees	Number of establishments on frame	Number of establishments selected for sample	Sample weight	Number of establishments responding
0-19	8	1000	50	20	30
20-99	35	500	50	10	35
100-249	175	200	50	4	40
250 or more	3,000	100	100	1	90

Calculate the following survey response rates using the formulas from the class notes:

- $$RR1 = \frac{I}{(I + P) + (R + NC + O) + U}$$
- $$RRW1 = \frac{\sum \omega_i I_i}{\sum \omega_i (I_i + P_i + R_i + NC_i + O_i + U_i)}$$
- $$RRMOSW1 = \frac{\sum \omega_i y_i I_i}{\sum \omega_i y_i (I_i + P_i + R_i + NC_i + O_i + U_i)}$$

What do each of these rates tell you?